A Hybrid Architecture for Low Latency AR Alignment

Matthew Desaulniers
Department of Computer Science
Colorado School of Mines
Golden, USA
mdesaulniers@mines.edu

Abstract

Multi-user Augmented Reality (AR) systems are critical for the enhancement of existing and future collaborative robotic systems. Multi-user systems provide information that may otherwise be unavailable to an independent user and augmented reality enables the integration of that information naturally into the human perception system. Multi-user AR systems can scale large, become complex, and generate quantities of data that must be managed and communicated between users. This paper describes efforts to lower human perceived end-to-end latency of a multi-user AR system. The system is measured with respect to user perceived latency as a motivation for directing efforts for future system-level research as well as improvements. We find that the latency bottleneck is associated with the cloud server based user alignment system. We augment the system with a peer-to-peer alignment scheme that runs in parallel to the cloud server based alignment. We implement the new user alignment system and measure latency and user alignment accuracy with respect to the user expected cloud server alignment. Finally we analyze data from experiments conducted with the fully implemented system. We find that the peer-to-peer hybrid system along with our proposed gravity-based alignment drastically reduces user perceived latency at the cost of some variance in positional and angular accuracy. We discuss future work on how the system may be made more robust to scaling and network conditions. latency

Index Terms: Augmented Reality, Point Registration, Alignment.

1 Introduction

As AR technology advances, its uses will expand into real-time applications. Scenarios like emergency response, surgical procedures, piloting aircraft, and vehicle operation will require extremely low latency in order to be considered feasible application spaces.

A promising benefit of these systems is the ability for more cohesive collaboration between humans and robots. A great advantage of human-robot collaboration is the ability to combine the the high-level decision making of a human along with the deep level of information that may be provided by sensors accessible to machines. One such application is for users to see occluded objects in real-time. For instance, suppose a warehouse forklift operator is navigating through aisles in a warehouse. There is no way of knowing if there may be another person or vehicle coming around any given corner without some form of secondary information besides auditory or basic visual cues. In this scenario, where a warehouse is already filled with modern sensors, like cameras, the perception ability of the forklift operator may be augmented by the warehouse sensor suite itself. Upon reaching an occluded turn, the operator is automatically given "line of sight" via rendered representations of people or obstacles that could not otherwise be seen. The process of giving a vehicle operator enough time to perceive and react to an incoming obstacle/person requires any solution to adhere to real-time latency constraints. This system is completely dependent on users having a shared reference frame such that obstacles appear where they would in the real world to give the user a sense of "line of sight" and must function with latency requirements on the scale of human reaction time.

These applications lend themselves to effectively extending the sensory abilities and perception of a user and so demand comparable latency to that of human reaction and sensing times. Achieving object placement and user to user alignment accurate enough for critical applications (cm to sub-cm scale) and strict adherence to low-latency requirements necessitate demanding design constraints for AR devices.

Achieving low latency in an AR system provides a unique challenge due to the applications in which it may be applied. From a technical perspective, many of the components that make up the system (SLAM, alignment, modeling, etc.) are well developed in their fields. These components work well on performance computing systems, but represent a challenge when the practical aspects of AR systems are applied. By nature of AR being a method of directly interacting with the world around a user, an AR system must be portable and small in size, whether with respect to vehicle or human. Not only must it be small in form factor, but it should also have high computational ability. These factors result in high cost for devices that render AR solutions infeasible within many application spaces. As in our previous warehouse scenario, the close proximity between the operator and object/person of interest requires the system to function at real-time latencies. This work aims to decrease end-to-end system latency without significantly sacrificing object placement accuracy using common off-the-shelf AR devices.

There are multiple perspectives in which this problem has been approached by the multi-user AR research community. Many works solve this trade-off between system cost and computational ability by offloading compute to edge or cloud servers. This decreases the local costs of computation on a user device, but increases total system complexity by introducing a network component and latency associated with network routing and physical distance between user and server. Conversely, some systems implement peer-to-peer solutions that take advantage of the close proximity of users to cooperatively attain better solutions or remove redundant calculations. Given the multi-faceted nature of AR systems, there are a number of research areas that have seen efforts to improve multi-user latency for streaming [7], SLAM [5], object detection [12] [4], and view alignment [9] [11]. For most state-of-the-art approaches, researchers tend to exclusively design either client-server based offloading systems or peer-to-peer systems. There is a much smaller subset of works that create hybrid systems utilizing a client-server model as well as a peer-to-peer component. These tend to lower end-to-end latency [8] [12].

User alignment is a fundamental component of multi-user AR

systems. Alignment ensures that users may interact with each other in a way that is spatially consistent across a group. User alignment is the process of finding the transformation between user reference frames such that we may correctly place objects in the same real-world position with respect to all users. An improperly aligned world view would result in one user seeing an object in a different position or orientation from another user. This is an essential, and often first, step that comes at the high cost of end-to-end system latency due to cloud servers having long network delay and latency due to computational overhead. We have set out to improve the peer-to-peer alignment latency of a cloud-based alignment anchor system. Our system creates a peer-to-peer alignment method that augments the existing client-server model. This new hybrid architecture will enable real time alignment that becomes more stable and consistent across sessions as the cloud server resolves its solution. The specific major contributions of this work are as follows.

- We have conducted an exploratory study on end-to-end latency of current solutions for user interactions and the data flow in multi-user AR of resource-constrained devices. This study led us to address the latency bottleneck in the cloud-based alignment process.
- We have propose a gravity-based user alignment approach that is able to resolve alignment an order of magnitude faster than cloud-based methods in resource-constrained mobile devices.
- We have implemented a hybrid architecture for multi-user AR that combines a client-server model and a peer-to-peer model using Android phones.
- We have compared the end-to-end latency of our system against a common cloud solution using Google Cloud servers. This resulted in reductions from over 3000ms to approximately 100ms at the expense of cm scale anchor jitter.

2 Related Work

User alignment in AR is generally accomplished in two ways.

First, alignment may be achieved as a byproduct of computing SLAM algorithms and communicating information relating to user maps [10]. This often involves either allowing an edge computer to host the centralized SLAM map, rendering the users dependent on the edge computer or allowing users to share maps between themselves, which can lead to complexity with keyframe selection and sending reduced size maps. SLAMShare [5] utilizes a nearby edge server to provide consistency for serving a hosted map to a group of users as well as reduced latency time from computational overhead. This drastically improves latency, but creates no method for alignment when an edge server is unavailable. FreeAR [1] effectively turns a peer-to-peer system into a client-server relationship, by electing one user to conduct all SLAM calculations for the group. The elected host device receives updates from peers and returns the peer position in its map. While this reduces some latency in the form of time taken to synchronize maps, the alignment is still only as accurate as a the ability of a mobile device will allow.

The second method is through direct user to user alignment. This has been achieved a number of ways. The concept of sending keyframes is a recurring theme in these works. A keyframe may be defined as a frame within a sequence of spatial data (this may be video or point clouds) that is chosen as a particularly useful frame. Usefulness may be due to a particularly rich feature set or the timing of the frame itself with respect to the system processes. In some cases, the keyframe is 3D data that is compressed back into a 2D data structure, like an image, and sent as if it were a feature-rich image frame [11]. Alternatively, some have used the keyframes generated from SLAM to begin a separate alignment process [9]. As SLAM keyframes are generated, they are selectively saved off until a number of keyframes have been accumulated. This is then sent as a message to peers to resolve. Keyframes are chosen by proximity to an object of interest that a host wishes peers to resolve with respect to.

All direct alignment works were found to be peer-to-peer based and works using SLAM to begin with were split between peer-to-peer and client-server based architectures. While these all lowered system latency in some way, the works that created peer-to-peer systems were left with systems that were not as accurate as client-server models and client-server models suffer from either network delay or leave no explanation for how latency may be affected when the server is unavailable.

SEAR [12] utilizes an architecture for object detection for multi-user AR that allows the system to take advantage of both peer-to-peer and client-server models. As in the previous server-based approaches, users may request object detection tasks from the edge server. In many cases the server will comply and return a result. Upon receiving the result the user will cache the response. As other users make requests, the server may deny a request. If this happens, users will request their peers to resolve it. All peers then check their caches and return the result if found. While SEAR simply brings two simple architectures together, it results in a massive gain for multi-user object detection. Likewise, SCAXR [8] implements a similar tiered architecture. An edge computer is assigned as the scheduler and this computer then monitors server resource use vs computational ability of users. If the scheduler needs to create room for server usage, it may direct a request to another user. The SCAXR system greatly improved rendering task latency for multi-user AR systems.

While there exist peer-to-peer alignment systems and client-server based alignment systems, there exists no hybrid architecture alignment systems that take advantage of the speed of peer-to-peer solutions and accuracy of server-based solutions.

3 Exploratory Studies

The purpose of this exploratory study is two-fold. The implementation of a multi-user AR application will yield a framework that can be used to evaluate existing AR features/algorithms and to identify aspects within these applications that may present themselves as weak points in current multi-user AR systems.

We implemented a multi-user AR system using three Google Pixel 5 smartphones. Each phone contains a Snapdragon 765G processor and 8GB of RAM. They are running on the Android 14 operating system. This model phone does not have LiDAR, but uses dual cameras for depth sensing. All three phones operate on the same local network.

We implemented a simple AR application where users make basic commands (i.e. place object, resolve cloud anchors, and host cloud anchors) that are propagated to other devices. The system is made up of user devices that are all peers, able to freely send information between themselves, and a cloud server. The role of the cloud server is to allow the users to conduct accurate world alignment between their respective reference frames. The cloud server hosts the original data used by a host user device to initiate an alignment as well as executes an alignment algorithm for any user that may request it. A host user device is simply the peer that initiates that request to host a cloud anchor and sends the cloud anchor ID to its peers. Peer devices that receive the ID and query the cloud server are referred to as a resolver device. Once user alignment is achieved, user devices may automatically recognize objects/images of interest in their environment and send their location to their peers. Those peers will then use the cloud-based alignment system to place those objects accurately within their own reference frame. The alignment allows users to place objects that other users will automatically see in a way that is spatially consistent. If some user places in object in a shared space, then all other

users in the space will see the object where that user placed it.

Application development was conducted in the widely-used Unity game development platform and AR-specific functions were built from AR Foundation and ARCore libraries. We used Google's cloud-based Cloud Anchor alignment system. UDP sockets were used for communication between peer devices and the ARCore API was used for making requests to Google Cloud servers. This application consists of several major components: SLAM, object detection, streaming and peer-to-peer communication, and alignment. We measured the latencies incurred by these components and show the results in Table 1.

Process	Time (ms)
Hosting Cloud Anchor (t_{host})	3,725
Host sends anchor ID to peers	6.2
Peer Receives Anchor ID	3.5
Peer resolves Cloud Anchor ($t_{resolve}$)	728
User communications AR object to peers	9.2
Peer places object locally	1.05

Table 1: Latency breakdown of exploratory system.

We first observe that the actual placement of objects takes a trivial amount of time (approximately 1ms). Any communication related processes also are low latency tasks, as every peer has connectivity to every other peer. We further observe that Hosting Cloud Anchor and Peer Resolves Cloud Anchor items are by far the largest latency bottleneck in the system. These processes are the points where a user has made a request to create an anchor that may be used for user alignment and when peer users try to resolve that alignment respectively. Given this massive disparity between user alignment latency and other functions, we believe the greatest benefits to user-perceived system latency is to reduce the latency of the cloud-based alignment system.

Figure 1 shows a more detailed view of the hosting and resolving process between the cloud server and users.

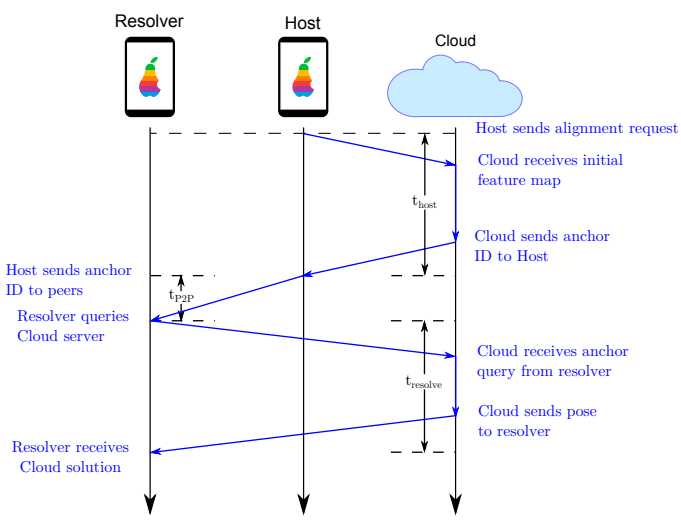

Figure 1: Benchmark AR system with users requesting Cloud Anchor hosting from Google servers (cloud).

4 Our Approach

To achieve a system that provides users with real-time alignment, we propose a hybrid architecture that uses peer-to-peer results to supply an almost instant user alignment. The real-time user alignment is then replaced by a higher latency, but more accurate cloud-based alignment. With this augmentation the overall system adds no new devices, but now the host and resolver devices have additional functionality. The host device, upon initiation of a request to the cloud server, now begins streaming its point cloud view to resolver devices. Resolver devices receive point clouds from the host to execute a local alignment algorithm until a cloud anchor ID is received from the host that the resolver may use to query the cloud server.

The two distinct approaches that allow this system to perform so much faster are its hybrid architecture and its gravity constrained point cloud alignment algorithm that runs extremely fast on resource-constrained user devices.

The hybrid architecture allows for the cloud-based alignment to be computed in parallel to the peer-to-peer user alignment system. While the host is awaiting an alignment ID from the cloud server, it begins streaming the associated point cloud of the cloud anchor location as well as its gravity vector to resolver user devices. The streamed point cloud is sparse, 50-200 points local to the selected location, so that the network and computationally latency may be minimized. Resolvers then take the gravity vector of both devices into account to run a user alignment algorithm at periodic intervals to generate new peer-to-peer alignment estimates. By running this algorithm in parallel, the same system used for exploratory studies may now facilitate user interaction within approximately 100ms.

5 Hybrid Architecture

The hybrid architecture may be viewed as a system that is accomplished by splitting up the user alignment into two steps. Once an alignment request with the cloud server is initiated by the host, the peer group first executes user alignment via peer-to-peer streaming of the host's point cloud to the resolvers. This is followed by the second step wherein the cloud server has responded to the request and peer users begin using the cloud-based user alignment solution.

To initiate an alignment request the host selects an area they would like to create as a reference point to communicate for peer alignment. Upon selection, the reference point and video frames are sent to a cloud server so a map of visual features (edges, corners, etc) may be created and hosted within the server to define the user requested reference point. This cloud request takes time to propagate through the network, be resolved, and return a lookup ID for the host to distribute to resolvers. After the host has selected the reference frame and awaits a response from the cloud server, it then begins to stream a lightweight point cloud to its peers with period $t_{streaming}$.

This presents a technical dilemma: when a user receives point cloud data in a peer-to-peer manner, how do they know whether their user alignment was accurate enough? Given the users are trying to establish a shared reference point, there is no method of doing this without human intervention or a previous reference point. In order to keep the application automated and experience immersive, a device cannot simply trust one point cloud alignment. One might conceivably allow for enough computation time that an accurate solution is guaranteed, but this would no longer satisfy low latency requirements. For this reason, point clouds are streamed at an interval such that resolvers may have opportunities to replace erroneous point cloud alignments when they come in.

This process is detailed in Figure 2 below.

5.1 Gravity-Based Alignment

Although AR devices are generally resource-constrained, they often are equipped with a suite of sensors that may be used to compensate for a lack of pure computational power. Specifically, IMU modules are ubiquitous in mobile devices and head mounted displays (HMD). By taking advantage of the IMU's ability to measure earth's gravity, we reduce the dimensions of the solution space from six to four. The peer-to-peer point registration algorithm uses

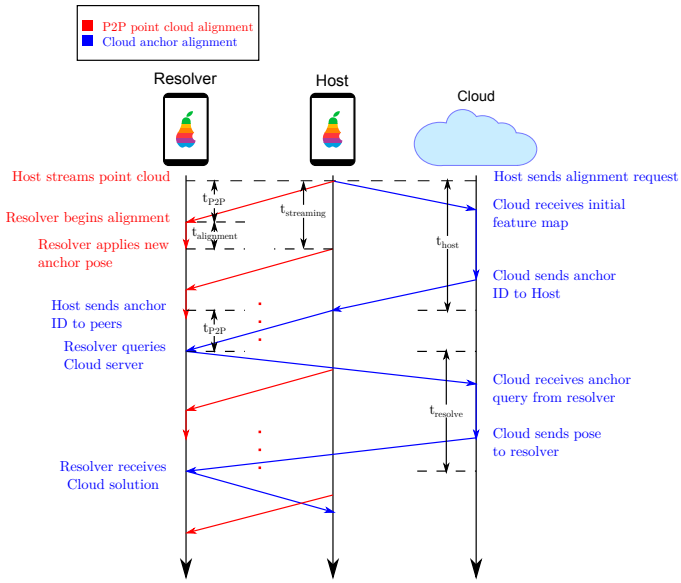

Figure 2: Hybrid system. The sequence of peer-to-peer streamed point cloud messages are shown in red. The cloud-based alignment is shown in blue.

a gravity-based user alignment strategy that can be broken up into three phases: pre-processing, finding an initial homogenous transformation guess, and finally using this as a starting point for an iterative closest point solver. Although all three phases are critical to finding an alignment between users, the use of a gravity vector is particularly unique to the system hardware and use case.

In summary, each peer filters their own point clouds and uses a gravity-based user alignment algorithm to select an initial estimate for an iterative closest point (ICP) algorithm. After a transform between point clouds is acquired, the views are aligned between host and resolver. This peer-to-peer process is periodically repeated. There is a balance to be found in designing this system between accuracy of resolve and $t_{alignment}$. $t_{alignment}$ cannot be so short that accuracy is unusable, yet also cannot be so long that it does not significantly improve the user perceived latency of the system. This balance must be met by satisfying the requirements demanded by our use case. This system must be able to provide input for humans and so should be responsive within approximately human reaction time, which is approximately 100ms [3]

Phase 1. Pre-processing: Our pre-processing steps were designed with our specific use case in mind. In this scenario, we have a group of users that are known to be within close proximity to each other. Not only do they share close proximity, but we also have some notion of group attention. Low latency would not be a requirement if the group was not trying to interact with some common point in space. We may use this information to filter out information outside of this shared point of interest (this will be the location of the alignment anchor). The resolving devices first conduct frustum culling to remove points outside of their view and a distance based filter that removes points that are outside of reasonable interaction distance from the viewer. The final step in preprocessing is gravity alignment.

A gravity vector is generated using the onboard IMU and filtered with a low pass filter,

$$g_{n+1} = (1-a)g_n + a\hat{g}_n \tag{1}$$

Where g_n is the estimated gravity vector at time step n and $\hat{g}_n$ is the accelerometer measurement at time n. Creating this estimate of gravity vectors allows the system to solve for the quaternion responsible for pitch and roll rotations from the host space to the resolver space.

This yields a new quaternion rotation from the resolver reference frame to host reference frame ${}^H\mathbf{q}_R$. Let the set of points defining the point cloud of the host and resolver be defined as such,

$${}^RP^R = \{{}^R\mathbf{p}_i^R \in \mathbb{R}^3 | i = 0,1,...,n\} \tag{2}$$

$${}^HP^H = \{{}^H\mathbf{p}_j^H \in \mathbb{R}^3 | j = 0,1,...,m\} \tag{3}$$

Where ${}^HP^H$ is the point cloud received from the Host device in the host reference frame and ${}^RP^R$ is the point cloud in the Resolving device reference frame. This rotation is then applied to the resolver point cloud,

$${}^HP^R = \{{}^H\mathbf{q}_R(0,{}^R\mathbf{p}^R){}^H\mathbf{q}_R^*|{}^R\mathbf{p}^R \in {}^RP^R\} \tag{4}$$

Phase 2: Initial Guess Our alignment system now has determined the roll and pitch angles between the host and resolver point clouds. Now that both point clouds are in the Host reference frame, the Resolver will run a 4DOF Random Sample Consensus (RANSAC) algorithm to find an initial guess at correspondences as well as Singular Value Decomposition (SVD) to find the initial transformation guess. RANSAC is a randomized algorithm commonly used for finding good fits for data points that is robust to outliers. RANSAC takes a small, sample of each data set, calculates some transformation between the two, and counts the number of inlier data points using this transformation. This process is repeated and the transformation with the largest number of inliers is chosen. The choice of transformation in our case is guided by physically rotating the point clouds in space and for that reason, SVD is a practical choice.

SVD is a method of calculating correlation of points in a dataset. Specifically SVD provides eigenvectors that represent the major and minor axes upon which the dataset it oriented. By creating a covariance matrix between RANSAC sample guesses, we may generate transformations that correlate (or physically align) the sample points in space. It is important to note that the SVD calculation blindly assumes that the point clouds have the same mean, so this implementation is sensitive to the success of preprocessing assumptions.

The initial guess that will be given to the ICP algorithm needs a yaw (ψ) value and position vector. This is generated by running a 4DOF RANSAC algorithm that matches linear features.

The overall execution of RANSAC will follow the traditional method [6]. The goal of creating the initial guess is to find the best location in the (ψ, x, y, z) search space that will allow the ICP algorithm to find the optimal alignment without getting stuck in a locally optimal solution. Due to the use of lightweight point clouds for fast computation, RANSAC must operate on a sparse feature set of approximately 25-100 $\frac{points}{m^2}$. This is a small RANSAC search space (40,000 possible correspondences) by many standards, but still an order of magnitude too large to rely on traditional point to point comparisons by constrained user devices.

To ensure good correspondences, we elect to compare visual features. In order to select candidate RANSAC features, the system selects two points to create a distance feature found in the data. This feature is the distance between two points such that it is larger than some threshold to avoid large yaw error. The resolver choose two points,

$$|{}^H\mathbf{p}_i^R - {}^H\mathbf{p}_j^R| = d_{ij}^R < d_{threshold} \tag{5}$$

Once a linear distance feature is found to satisfy this constraint it is used to find a corresponding feature in the host data set such that the features are the same distance within some small error.

$$|{}^{H}\mathbf{p}_i^H - {}^{H}\mathbf{p}_j^H| = d_{ij}^R < d_{threshold} \tag{6}$$

$$|d_{ij}^R - d_{ij}^H| < d_{error} \tag{7}$$

If this constraint is satisfied then the algorithm assumes a match and the traditional inlier count and SVD solution is executed to find an initial guess of the 4DOF transformation is returned.

Phase 3: Transformation Solver

Solving for the transformation is accomplished by running an ICP solver [2]. This is accomplished using a simple closest point correspondence metric such that points are paired with their nearest neighbors such that they are no greater than 10cm away to filter out outliers. Due to the preprocessing and initial guess steps completed, we expect that the ICP algorithm will be provided with point clouds with parallel x-z planes and similar centers of mass that only need to be translated and rotated about the y axis. This will allow the algorithm solution to iteratively find the local minima near the initial guess.

6 Experimental Results

Experiments were conducted in a variety of well-lit indoor environments. The system begins streaming once the host device makes a request query to the cloud server and is awaiting a response. We find that the average latency is 119.69ms for each peer-to-peer user alignment. This effectively reduces the user perceived latency of the multi-user system from 3.725 seconds to 119.69ms or an approximate 97% reduction in latency. Table 2 shows the functions of the benchmark system now including the parallel host transmission of point cloud data and resolver time. To gather results, 34 requests to host a cloud anchor were made to the Google cloud server that resulted in 392 peer-to-peer user alignment calculations. Network conditions and varying wait times for the response from the Google cloud server means that the number of user alignments per hosting event is variable. The average number of alignments per host event is 11.5.

Process	Time (ms)
Hosting Cloud Anchor (t_{host})	3,684
Host sends point cloud data	6
Resolver solves alignment ($t_{alignment}$)	119.69
Host sends anchor ID to peers	6.2
Peer Receives Anchor ID	3.5
Peer resolves Cloud Anchor ($t_{resolve}$)	922
User communications AR object to peers	9.2
Peer places object locally	1.05

Table 2: Latency breakdown of all processes included in the hybrid architecture.

This reduction in latency comes with a less spatially accurate alignment between users. Figure 3 below shows the error distribution in x, y, and z directions respectively. Error is measured with respect to the cloud served alignment solution.

The average error, seen in Figure 4, in the x, y, and z directions is -0.2133cm, 0.5718cm, and -1.0175cm respectively. Although this shows that the algorithm is on average accurate within a cm, it is clear that there is a notable variance about the average for every direction. One feature of note is the lower variance in the y direction. Unity uses the left hand coordinate convention with the y axis facing up, so this corresponds the the axis about which the yaw values are being solved. It follows that this is the axis with the tightest spread in variance as it represents the roll and pitch locked plane "lifting" away from corresponding point cloud plane and quickly grows in error as a solution candidate.

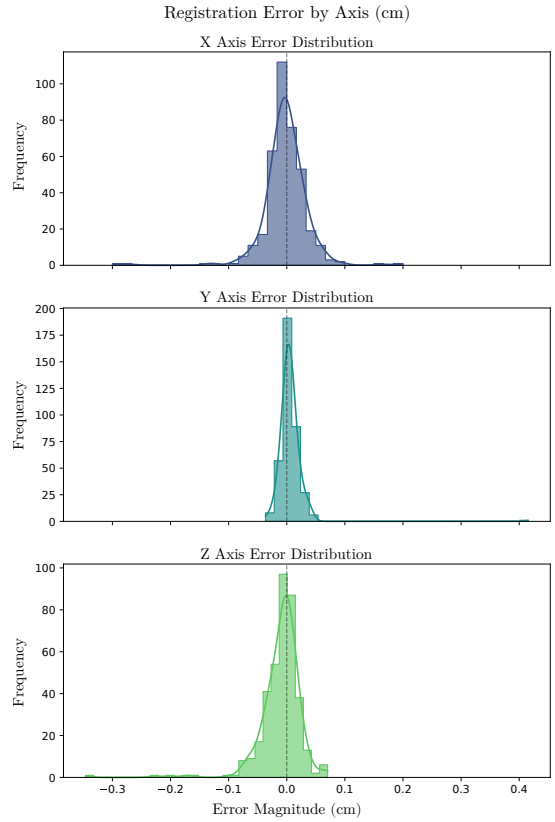

Figure 3: Distribution of linear error with respect to cloud solution. (Top) x error in cm. (Middle) y error in cm. (Bottom) z error in cm.

Figure 5 illustrates the variation of angular accuracy resolution. Note the roll and pitch do not share the same x axis scale as the yaw data. This shows an extremely tight grouping when compared to yaw values due to the use of IMU gravity alignment to solve for roll and pitch values directly.

Looking at the mean and standard deviation in Figure 6 reveals the variance rotation accuracy about all three axes. As expected, roll and pitch errors are extremely small on average and do not vary greatly outside of one degree. In the current system we see a large variance in yaw values. With outliers in Figure 6 as large as $-170°$ to $170°$ and values within one standard deviation fitting between $-19°$ and $19.75°$. Due to the cloud server being used as a reference point for error, there is some chance that the roll and pitch values solved for by the peer-to-peer system are actually more accurate than the cloud based system and these errors are a manifestation of the limitations of the cloud-based alignment system.

7 Conclusions

From the perspective of data analysis, there are extensions to the system that would greatly enhance the ability to extract information from the system. First, the data collected was measured with respect to the cloud server provided alignment. This means that we can see how smooth the transition from peer-to-peer solution to the cloud solution may look for the user. Alternatively, creating an absolute reference point that would automatically place an anchor at a known world location would allow independent errors to be collected from both solutions. This may provide insight into where peer-to-peer solutions may lead to better results than cloud solutions. The second extension would be to use the transform between the world frame and user camera frame to transform points before conducting data analysis. The mean error in Section 6 may be cor-

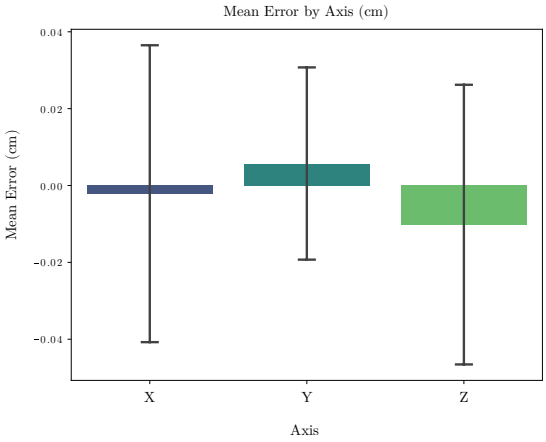

Figure 4: Average linear error in each linear direction with respect to the cloud server solution.

related with the direction of the vector orthogonal to the camera plane, as cameras have less accurate tracking depth. If these were quantified in our system, it could be used to create a more accurate algorithm that compensates for this effect.

Future work includes further connecting the greater system design in ways that make it more robust to network reliability. A new user may enter the interaction that was not within the network at the time of the initial resolve routine if the cloud server is no longer available, the user may request point cloud feature data from *more than one* user in the group. This would allow the user to create a much more accurate alignment estimation than the process described above by combining the results of multiple alignments. While this would take longer, it would serve as a better permanent anchor when there are users available, but no cloud server. From a single user's perspective, this algorithm could be improved by the addition of some kind of tracked state. Once an alignment is solved for, this could be used as the beginning of an estimation algorithm that updates the state of the alignment rather than simply replacing it as the system currently does. This idea was attempted, but with no promising results in the time available.

Scaling in this system is linear in most functions. AR object placement is conducted in a complete network and so a placement from one user to n users only takes n messages at most. Alignment for both peer-to-peer and cloud algorithms work the same way. For this reason, it would be necessary to test the absolute limits of scalability using a large number of devices. Future work would further include using docker to simulate devices sending synthetic data to test this level of scalability.

The user alignment system created here serves the purpose of reducing user-perceived end-to-end system latency. In a collaborative robotics scenario such as transportation, search and rescue, or our aforementioned forklift operator, users must be provided accurate information that they may use to plan and receive it fast enough that the information may be acted upon. The linear scaling of this system allows any peer to become a host that may seed the alignment. This alignment may then be used by any number of users to send/place AR objects or information to an number of users in a decentralized fashion. Although future work would enhance the robustness of the alignment and quantify the limits of user scaling, this work accomplishes feasibility for fast peer-to-peer alignment, accurate cloud alignment and user interactions are not scale limited by system architecture.

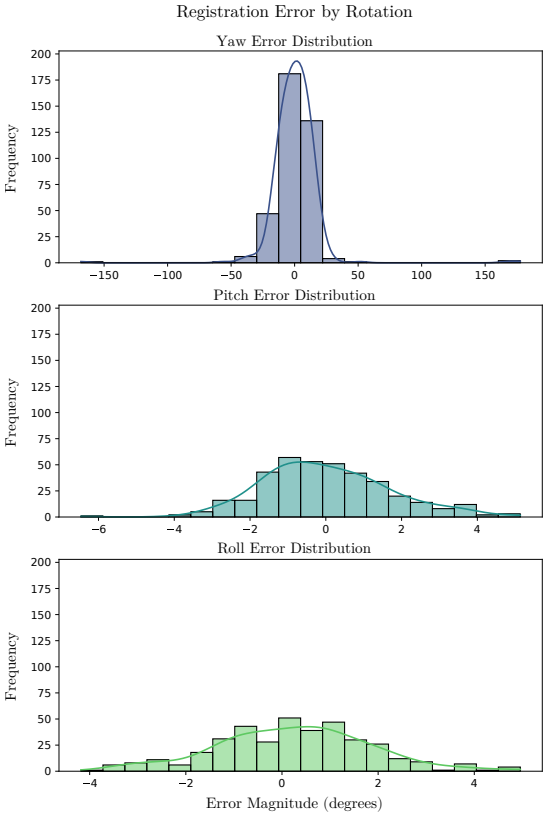

Figure 5: Distribution of angular error about each linear axis. (Top) Roll error in degrees. (Middle) Yaw error in degrees. (Bottom) Pitch error in degrees.

References

[1] K. Apicharttrisorn, J. Chen, V. Sekar, A. Rowe, and S. V. Krishnamurthy. Breaking edge shackles: Infrastructure-free collaborative mobile augmented reality. In *proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems (SenSys)*, p. 1–15, 2023. 2

[2] P. Besl and N. D. McKay. A method for registration of 3-d shapes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 14(2):239–256, 1992. doi: 10.1109/34.121791 5

[3] K. Chen, T. Li, H.-S. Kim, D. E. Culler, and R. H. Katz. Marvel: Enabling mobile augmented reality with low energy and low latency. In *proceedings of the 16th ACM Conference on Embedded Networked Sensor Systems (SenSys)*, p. 292–304, 2018. 4

[4] V. Cozzolino, L. Tonetto, N. Mohan, A. Y. Ding, and J. Ott. Nimbus: Towards latency-energy efficient task offloading for ar services. *IEEE Transactions on Cloud Computing*, 11(2):1530–1545, 2023. 1

[5] A. Dhakal, X. Ran, Y. Wang, J. Chen, and K. K. Ramakrishnan. Slamshare: visual simultaneous localization and mapping for real-time multi-user augmented reality. In *proceedings of the 18th International Conference on Emerging Networking EXperiments and Technologies (CoNEXT)*, p. 293–306, 2022. 1, 2

[6] M. A. Fischler and R. C. Bolles. Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Commun. ACM*, 24(6):381–395, June 1981. doi: 10.1145/358669.358692 4

[7] Y. Han, Y. Chen, R. Wang, J. Wu, and M. Gorlatova. Intelli-ar preloading: A learning approach to proactive hologram transmissions in mobile ar. *IEEE Internet of Things Journal*, 9(18):17714–17727, 2022. 1

[8] Y. Huang, H. Wang, X. Qiao, X. Su, Y. Li, S. Dustdar, and P. Zhang. Scaxr: Empowering scalable multi-user interaction for heterogeneous xr devices. *IEEE Network*, 38(4):250–258, 2024. 1, 2

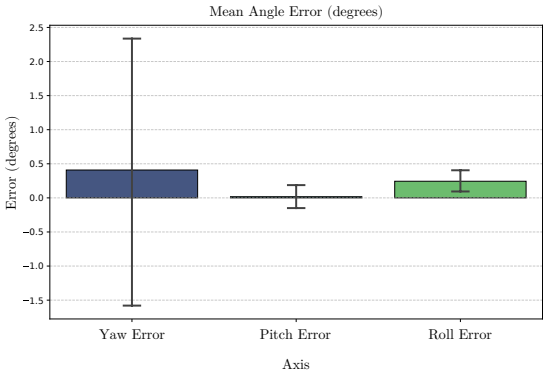

Figure 6: Average angular error about each axis.

[9] X. Ran, C. Slocum, Y.-Z. Tsai, K. Apicharttrisorn, M. Gorlatova, and J. Chen. Multi-user augmented reality with communication efficient and spatially consistent virtual objects. In *proceedings of the 16th International Conference on Emerging Networking EXperiments and Technologies (CoNEXT)*, p. 386–398, 2020. 1, 2

[10] K. Sartipi, R. C. DuToit, C. B. Cobar, and S. I. Roumeliotis. Decentralized visual-inertial localization and mapping on mobile devices for augmented reality. In *proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2145–2152, 2019. 2

[11] Y. Wen, A. B. Shah, R. Cao, C. Zhang, J. Chen, X. Wu, C. Xie, and X. Fu. Ar-light: Enabling fast and lightweight multi-user augmented reality via semantic segmentation and collaborative view synchronization. *IEEE Transactions on Computers*, 74(6):2073–2086, 2025. 1, 2

[12] W. Zhang, B. Han, and P. Hui. Sear: Scaling experiences in multi-user augmented reality. *IEEE Transactions on Visualization and Computer Graphics*, 28(5):1982–1992, 2022. 1, 2